

Intention forming and processing of information about desirability and importance

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Empirical evidence suggests that predecisional information processing is biased in favor of an emerging preference resulting in coherence effects (Glöckner & Betsch, 2008; Simon & Holyoak, 2002; Simon, Pham, Le, & Holyoak, 2001). After the decision coherence effects have been shown to fade away (Simon, Krawczyk, Bleicher, & Holyoak, 2008). This transience of coherence effects seems to conflict with empirical findings supporting mindset theory (Gollwitzer, 1990, 1991), which describes processes of action implementation that come after the decision. Here, biased postdecisional processing of information in favor of the intention is understood as one of the main characteristics of successful goal attainment. In two studies we explored explanations for this divergence and possible links between the two approaches (Hiemisch, Meyer-Gerdes & Pohlen, in prep).

The presented results imply that the two approaches are not mutually exclusive, but rather complementary. The processes of coherence construction flexibly reacts to situational task demands (Freeman & Ambady, 2011), which might change between the process of making the decision and the process of implementing relevant actions. Predecisionally, biased processing enables decisions in spite of conflicting information, whereby a relative openness for cons prevents that problems are completely ignored. In contrast, postdecisionally a shielding against cons supports intention behavior stability. Hence, transience of coherence effects can be expected to be observed in situation in which participants make decisions without implementing them.

In the second part of the presentation we will discuss the design of a study, planned to trace the predecisional processing of the desirability of cue values and the importance of cues for different types of decisions. Specifically, we compare decisions people have to implement with decisions that cease to be of interest after they were made.

The antecedents of escalation behavior: How to differentiate preference and initial decision?

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First, we present the results of two studies that related important dimensions of escalation research: Sunk costs and responsibility. The sunk cost effect describes the tendency to invest additional resources in a losing course of action because of previous investments, while responsibility for the initial decision leads to higher investments in the face of negative feedback

Approaches of postdecisional bias (e.g. Festinger, 1957; Gollwitzer, 1990) assume that the initial decision for a project causes a bias in its favor that result in reduced processing of postdecisional information (e.g. negative feedback, sunk cost). In contrast, Schulz-Hardt, Thurow-Kröning and Frey (2009) argue that escalation is first and foremost driven by preferences previous to the initial decision. In their studies they showed that the responsibility effect was completely mediated by these preferences.

In Study 1, participants who made an initial decision showed no sunk cost effect. In contrast, a large sunk cost effect was found for non-responsible participants. In Study 2, all participants were asked to report their preferences. The sunk cost had no effect on further investments. Investments depended on the preferences, and responsibility explained no additional variance.

The results raise a series of questions. Is a reported preference in an escalation scenario a kind of initial decision (decision framing) that inhibits the sunk cost effect? Or is the development of a preference sufficient to inhibit the sunk cost effect? Therefore, in the second part of the presentation we will present and discuss an empirical design that is planned to trace the processes of developing a preference, making an initial decision about a course of action and implement the course of action.

Distortion of moral judgments in a decisions under temptation

Tadeusz Tyszka

Koźminski University, Poland

The hypothesis was tested that when a choice alternative is profitable but at the same time unethical, the decision maker will tend to neutralize its moral disadvantage through reformulating it in a more neutral or positive way. One method to achieve this goal (perhaps, when the decision maker follows a lexicographic decision rule) is to categorize the act in such a way that the category label no longer evokes any strong negative moral emotions. An alternative way (perhaps when the decision maker follows a compensatory decision rule) is to distort the moral negative outcome of the act and its probability.

The Role of Numeracy and Emotion in Health Risk Perception and Decision Making

**Andrea Taylor and Rob Ranyard
University of Bolton, UK**

In recent years numeracy has been increasingly recognised as a factor that can influence both risk comprehension and choice process. In a previous series of studies we investigated the relationship between numeracy and adherence to rational choice principles in decisions under risk. Both decision behaviour and process data (reaction times and think aloud protocols) suggested that as decision complexity increased, less numerate individuals became increasingly likely to neglect probabilistic information. While these findings are compelling they were observed in the clearly defined and affectively pallid context of the lottery choice paradigm. Consequently, our findings may not fully capture the relationship between numeracy and information processing in more emotionally salient choice scenarios and those where information must be actively sought. We therefore propose three studies which aim to investigate: a) the extent to which more and less numerate individuals utilise probabilistic information in hypothetical treatment scenarios varying in emotional salience and complexity; b) the manner in which more and less numerate individuals seek out probabilistic information in scenarios that require active information search; and c) whether manipulating information format in decisions under description can attenuate probability neglect.

Pro and con arguments in choices between psychotherapeutic treatment models

Marianne Jakobsson & Ola Svenson
Stockholm University, Sweden

The presentation focuses on decision process of psychology students who choose therapeutic training based on Cognitive Behavioural Therapy (CBT) or based on Psychodynamic Therapy (PDT) as a 1,5 years long part time course at the end of their training as psychologists.

Arguments were given by the psychologists via verbal protocols both for and against the chosen and the non-chosen option, before and after their decisions (4 sessions in total). Here, we focus on the contents of the arguments and the psychologists' estimates of the strength of their arguments pro and con at each session in a combination of qualitative and quantitative methods. The arguments were divided into categories and the categories were validated in computations of inter-rater reliability. Each category will be analysed in some detail using judgments of strength to see how the decision process can be traced in changes of the arguments/reasons pro and con before and after a choice.

Applying process tracing methods to study doctors' diagnostic reasoning

**Olga Kostopoulou, Miro Sirota, Andrea Rosen
King's College London, UK**

We will present two ongoing studies that employ process-tracing methodologies to investigate cognitive aspects of medical diagnosis. In addition to providing answers to applied problems, the research poses interesting questions about process tracing methodologies and models of the diagnostic process.

The first study investigates the effectiveness of two generic approaches to computerised diagnostic support: 1) suggesting diagnoses to doctors at the start of the encounter and with minimal patient information and 2) alerting doctors about diagnoses that they may have missed at the end of the encounter. To collect the data, a computer simulation was created that allowed us to trace the information seeking of each participant, as they diagnosed 9 patient scenarios, designed at different degrees of difficulty. Preliminary analyses based on 182 participants (final N=300) confirm the hypothesis that early suggestions are more effective than either late alerts or no support. We also plan to analyse the proportion of participants' information search that was related to the correct diagnosis and determine 1) whether it predicted final diagnostic accuracy and 2) whether early diagnostic support influenced subsequent information search. Discussion will centre around other behavioural variables that could be usefully investigated.

The second project is investigating why General Practitioners may miss early presentations of cancer. Early cancers present with non-specific features, which both patients and their doctors may attribute to other, more common conditions. We aim to 1) determine conditions under which the cancer diagnosis is less likely to be considered and 2) establish at which stage in the process the cancer diagnosis tends to get missed (initial hypothesis activation or subsequent hypothesis testing). For the first aim, we will conduct an online vignette study where, in a 2 x 2 factorial design, we manipulate disease prevalence and the presence of an alternative explanation for the patient's symptoms. For the second aim, we will conduct a process-tracing study, similar to the one described earlier. Participants will also be asked to verbalise their hypotheses at different stages in the process and to think aloud during some scenarios. Discussion will centre around other, non-invasive ways for checking hypothesis activation in medical diagnosis, without the need for overt verbalisations.

Using reaction times to investigate the generation of diagnostic hypotheses in medicine

York Hagmayer & Olga Kostopoulou,
King's College London, UK

Background: Clinicians have been found to miss diagnoses especially when a condition is rare or presents in a non-specific way. The diagnosis of early cancers in primary care is a prime example. This is not due to lack of knowledge, as the respective diagnoses and their associated signs and symptoms are known to clinicians. Hence, it is most likely due to a failure in activating the right hypotheses. Such activation processes in memory can be studied by measuring reaction times. If the observations made by a clinician trigger a hypothesis, reactions to terms related to the hypothesis should be accelerated relative to reactions to unrelated terms. The big advantage of using reaction times is that implicit activation of hypotheses can be studied before a hypothesis passes the activation threshold and reaches consciousness.

Aim: We will present and discuss research paradigms, which allow the study of hypothesis generation by means of reaction times. An ideal paradigm will be presented, in which the activation of several hypotheses after each cue is measured and compared to both explicit and implicit diagnostic hypotheses at the end of the information input. It will be shown, however, that this paradigm is not feasible for studies with clinicians as the number of required cases is too large. Two simpler alternatives will be presented and their limitations will be outlined. Through discussion with the audience, we hope to get additional ideas about a research paradigm that may work for doing real world research.

Framing of decisions: Effect on active and passive risk avoidance

Oswald Huber, Odilo W. Huber, Arlette S. Bär
University of Fribourg, Switzerland

If decision makers intend to avoid the risk in a risky decision situation, they can choose a less risky alternative (passive risk avoidance) or intervene actively in an alternative with a risk defusing operator (RDO). Up to now, only **negative** RDOs (preventing or compensating undesirable outcomes) have been investigated.

Positive RDOs as a complement to negative RDOs may also be possible in situations with an uncertain positive outcome. For example, a person examining the alternative to accept a job offer with the chance to get promoted within one year may reflect on possibilities to improve her chances, for example, by working very hard or by spinning intrigues against rivals.

We predict search for negative RDOs to be more frequent than that for positive RDOs. People are more ready to act in the face of a threat. Usually it is more urgent to eliminate a looming negative outcome than to promote a positive one.

We presented two scenarios to 64 participants, each including a safe alternative (sure positive and negative outcomes), and a risky one. The frame of the risky alternative was varied: In the positive frame, an uncertain event could lead to an improvement, in the negative frame to a worsening (logically equivalent frames). By using identical outcomes in both frames, we avoid confounding the frames with the attractiveness of alternatives.

As expected, more RDOs (indicating risk aversion) were searched for in the negative frame than in the positive one. This result contradicts the classical framing pattern. However, in cases without RDO (no search or unsuccessful search) we observed the standard result: risk-aversion in the positive frame and risk seeking in the negative frame. This result is remarkable, because only few participants searched for probabilities.

Invited Talk

Friday 4th May 2012

Speaker – Andreas Glöckner

Awaiting abstract

**Frontiers in Cognitive Process Tracing and Psychometrics:
development and Reverse Engineering of Cognitive Tests**

**Edward Cokeley
Michigan Technological University**

Awaiting abstract

The Cognitive Processes Underlying Risky Choice

Eduard Brandstätter
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Johannes Kepler University of Linz, Austria

In this article we investigated the cognitive processes underlying risky choice. We tested two concepts of decision making—heuristic choice and modifications of expected utility theory. We reviewed previous process tests of the priority heuristic, highlighted the restricted range of research methods used for those tests, and employed verbal data—think-aloud protocols and the newly proposed predict-aloud protocols—to overcome this lack of research methods. Our results are as follows: First, processes akin to those implied by the priority heuristic better captured violations of expected utility theory, including the possibility effect, the common ratio effect, the common consequence effect, the reflection effect, and the fourfold-pattern. Second, findings generally support the Heuristic Hypothesis, which embraces the idea that people examine information between rather than within gambles. Third, for easy problems (i.e., ratio of expected values is larger than 2) we found that people check for similarities across gambles to select the gamble having the higher expected value. Fourth, we introduced a new process-tracing technique, called predict-aloud protocols, and found converging evidence with think-aloud protocols when applied to the same set of problems. Although tests of the priority heuristic have been published before, the studies herein are the first using verbal data, including the newly proposed predict-aloud protocols. We further argue that predict-aloud protocols have advantages over classical think-aloud protocols and over previously suggested research methods. Together our findings suggest that cognitive operations similar to those implied by the priority heuristic best describe people's choice processes when problems are difficult, and people seem to rely on similarity processes when problems are easy.

Systemic cognition: Epistemological, conceptual, and methodological challenges.

Gaëlle Villejoubert
Kingston University London, UK

Process theory applied to judgements and decisions aims to investigate how people acquire and use mental representations of external events. Its methods encompass various techniques for tracing the steps people may follow in their mind to produce a judgement or reach a decision. A tacit assumption of this approach—and, more generally, of cognitive science—is that the information present outside the mind is first converted into a symbolic representation before being transformed by the mind and, finally, supplied to the outside world in the form of a decision or a judgment outcome.

An alternative view consists in positing that cognitive processes, or the mechanisms by which knowledge is acquired and used, may take place in a system that encompasses both symbolic representations inside the mind of the decision-maker as well as physical presentations of information outside her mind, in her immediate environment. The aim of this talk is to illustrate and discuss three challenges offered by this systemic view of cognition, based on recent and ongoing studies from our lab.

The first challenge is epistemological. Adopting a systemic view of cognition requires cognitive scientists to relinquish their current mindset and step back to redefine their focus of study as including a) the physical tools and material presentations available in the decision-maker's immediate environment, b) the mechanical actions that can be performed on these presentations alongside c) the mind's "black box". The second challenge is conceptual; adopting a systemic view of cognition calls for a redefinition of the cognitive objects that need to be traced as either internal representations or as external presentations. Similarly, two classes of cognitive processes should be distinguished: those involving a series of mental transformations of abstract representations and those involving a series of concrete transformations of material presentations. The last challenge is methodological. Current process tracing methods are designed to monitor internal representations and cognitive processes as they unfold in the mind; but can these methods be applied to study cognitive events taking place in the physical world or do we need to develop new tools?

Rationality and psychological accuracy of risky choice models based on option- vs. dimension-wise evaluations

Joanna Sokolowska and Zuzanna Obrebska
Warsaw School of Social Psychology, Poland

Two experiments were carried out to investigate psychological accuracy of risky choice models related either to classic or bounded rationality assumption. In line with the former, individuals trade-off outcomes and probabilities and choose the option with the highest weighted sum. Cumulative Prospect Theory (CPT) represents such models. In contrast, in models based on bounded rationality assumption, like Priority Heuristic (PH) or Maximization of Cumulated Probability of Winning (MCPW), there is neither trade-off between payoffs and probabilities nor complex information integration, since choice is based on dimension-wise comparisons. The latter models differ in emphasizing the significance of either the amount or the probability of loss. Previous findings do not clearly support either hierarchy.

Such hierarchy was tested in Experiment 1, in which 208 respondents chose one of four 2-mixed-outcome options. In Experiment 2 the relative accuracy of compensatory models based on global evaluations and non-compensatory models based on dimension-wise comparisons. The focus is on the way of reasoning represented by certain classes of models rather than on distinguishing between specific models with a given class (e.g. PH vs. MINIMAX). In This experiment subjects made choices among multi-outcome lotteries with gains and losses. Non-trivial multi-outcome lotteries with gains and losses provide stronger process tracing arguments than two-outcome lotteries. Information acquisition patterns were registered using a modified version of Mouselab Web. Choice fraction and process tracing indices were analyzed.

It was found that respondents made more dimension- than option-wise transitions. However, other indices such as the fraction of information searched across options as well as the ratio of searched information about pay-offs and probabilities do not support either model. The choice strategies used by a majority of respondents varied across situations.

Keywords: compensatory and non-compensatory models, information search patterns

Concurrent verbal reporting during information search: reactivity, completeness and limitations

**Thomas Scherndl & Anton Kühberger
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University of Salzburg, Austria**

Process tracing is a valuable source of data on judgment and decision processes. Consequently, JDM research has accumulated a toolkit of different process tracing techniques. Although there are many researchers arguing in favor of doing multiple process tracing, there is little research on the utility of applying multiple process tracing techniques concurrently. We conducted a study that compares the results of two classic process tracing techniques: concurrent thinking-aloud and information search patterns traced with MouselabWEB. We found that thinking-aloud led to longer decision times, although the think-aloud protocols were incomplete in comparison to the Mouselab data. More importantly, we also found different patterns of information acquisition and of information integration for Mouselab and the think-aloud procedure. Indeed, the agreement between these two techniques was poor indicating that the conclusions that can be drawn for the nature of human JDM depend on the method used. We discuss the implications of the (in) compatibility of these process tracing techniques for progress in JDM research in terms of reactivity and completeness, and propose questions for future research.

**Process tracing - quo vadis?
Reflections on a research approach and a glimpse at its future.**

**Michael Schulte-Mecklenbeck
Center for Cognitive and Decision Sciences, University of Basel
Swiss Federal Institute of Technology**

In this talk I will take a look at the history of different process tracing techniques and their contribution to our understanding of information acquisition processes. How has usage of different techniques developed over the years and which shifts can be observed. I will outline strategies to tackle problems such as method selection, data handling and data visualization. In addition to looking into established techniques, I want to highlight new developments such as Flashlight and Mousetracking which allow for new perspectives on processes of human cognition.

Moral intuitions of young Polish liberals and conservatives

Anna Macko

Kozminski University, Poland

One of the basic classifications of the political orientation is a division of the political spectrum into liberals and conservatives. According to the research by Haidt et al. (2007,2009) liberals and conservatives differ in regard to the issues (concepts, behaviours, etc.) they consider moral. The moral spectrum of the liberals is limited to the issues concerning the protection of individual rights, while for conservatives morality is mainly (but not exclusively) a question of issues that bind people into groups and institutions. However, such results are obtained in the countries with longer experience of democracy. Thus, it would be interesting to see whether the same mechanism holds also in Poland. The reported study aimed at answering the following questions: 1. Do members of different youth organizations associated with major political parties in Poland – young Polish liberals and conservatives – actually differ on the liberal-conservative dimension? 2. To what extent does the moral sphere of young people in Poland who describe themselves as liberals or conservatives resemble that of liberals and conservatives from the countries with a long tradition of democracy? 3. How far are the people with different political orientations able to reconstruct the moral reasoning of their ideological opponents? The results show that there are differences between members of various political youth organizations on the liberal-conservative dimension and significant similarities of liberals' and conservatives' moral spheres in Poland and in the countries with longer democracy experience. The results also show that there are differences between liberals and conservatives in reconstructing the moral reasoning of their ideological opponents.

Posters

Andrea Taylor, University of Bolton, UK

Awaiting abstract

Posters

Danielle Winstanley – University of Bolton, UK

Awaiting abstract

Skin conductance response (SCR) measurement in two sequential risk-taking tasks

Rebecca Wright
University of Essex

The Iowa Gambling Task (IGT, Bechara et al., 1994) and the Balloon Analogue Risk Task (BART, Lejuez et al., 2002) are sequential risk-taking tasks. The “standard” theoretical account of learning in the IGT draws heavily on skin conductance responses (SCR) data but relies on scarce SCR data and has attracted criticism (Dunn, Dalgleish & Lawrence, 2006). The BART has not previously been examined using SCRs. Consequently, SCRs were measured for both tasks to aid the understanding of implicit decision processes (Figner & Murphy, 2011). In the IGT, participants earn money by selecting cards from four unlabelled decks: two “good” decks containing low rewards and different frequencies of modest punishments, and two “bad” decks containing higher rewards and different frequencies of severe punishments. The BART requires participants to pump a computerised “balloon” to earn points. If a balloon reaches its individual burst point (known only to the experimenter) before a bank is made, the current points are lost. Study 1: Participants completed a computerised IGT. Conscious task knowledge was assessed using questions (Maia & McClelland, 2004). Just under half the participants were behaving advantageously according to their conscious awareness as revealed by their answers. Participants preferred the low frequency high magnitude punishment structure decks (bad or good) by the end of the game. Outcome SCRs increased with punishment magnitude; but, contrary to Bechara et al. (1994), participants did not develop anticipatory SCRs for the bad decks. Study 2: Participants completed the BART and IGT (without probe questions). Participants favoured the good IGT decks more strongly than in Study 1. SCR data analysis is ongoing, and will include comparison of SCRs following a bank or burst outcome in the BART. Future work will explore additional physiological measures, such as pupil dilation to assess emotional reactions and mental effort in sequential risk-taking tasks.